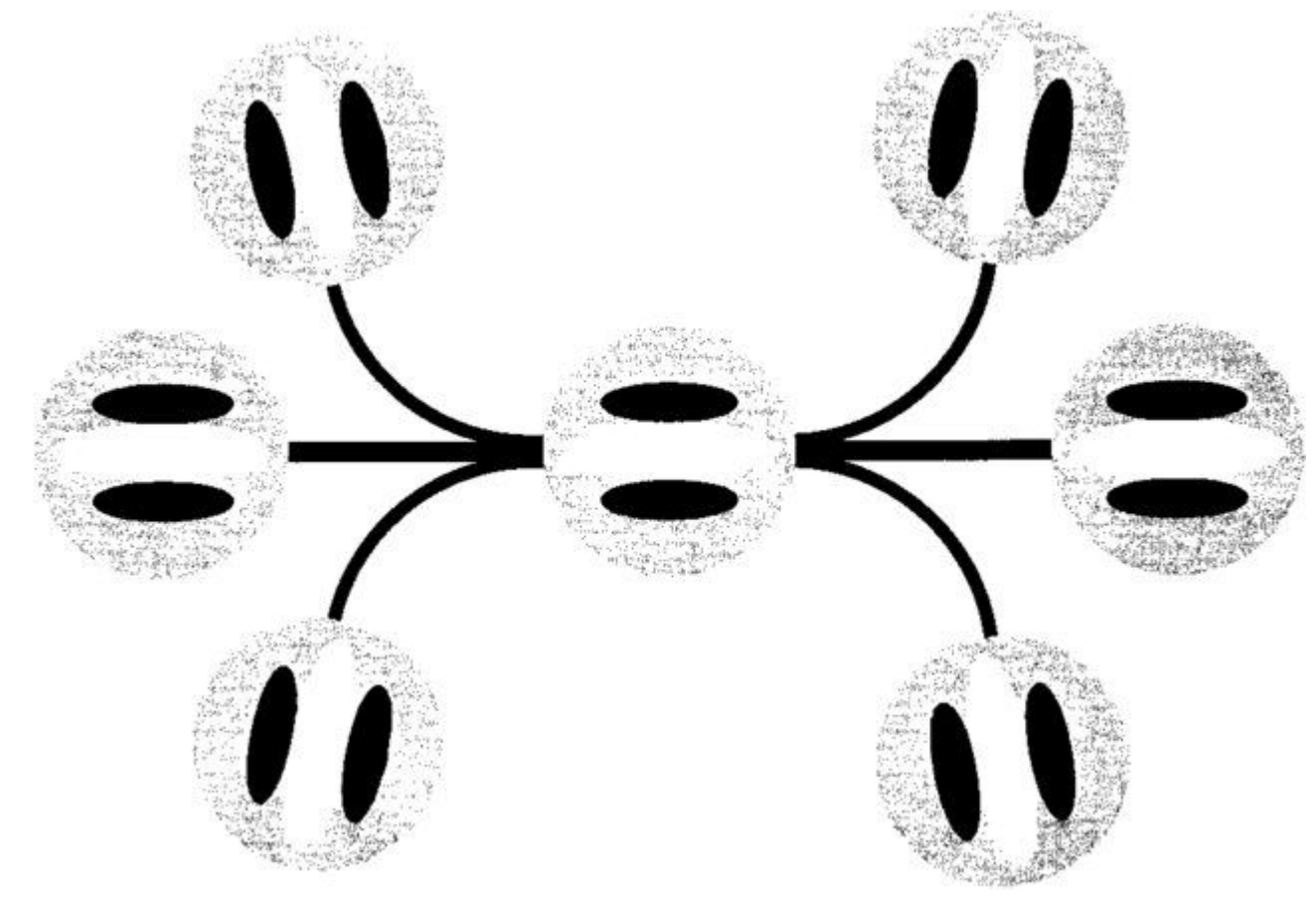


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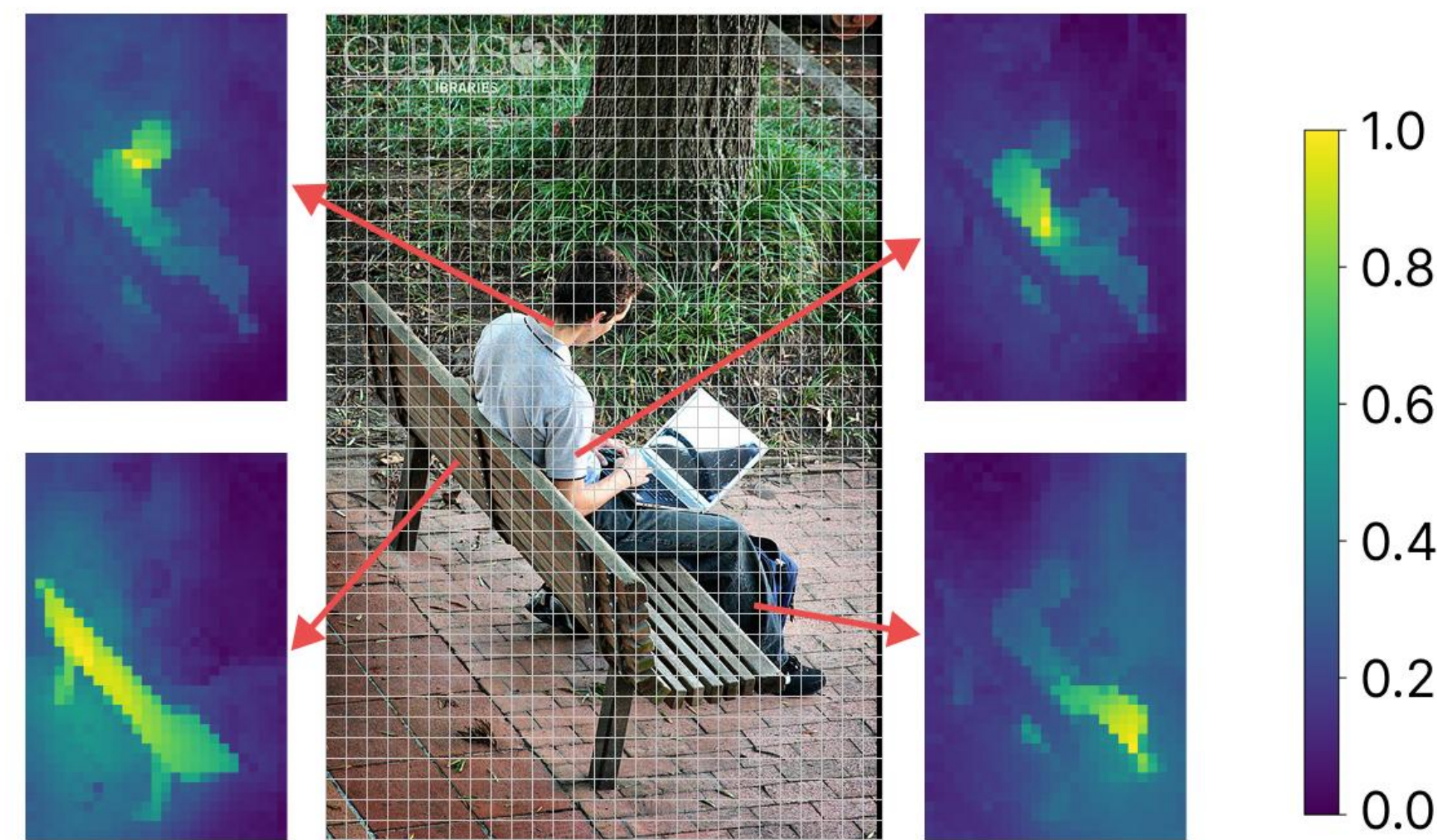
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Association Fields in V1



Field et al, (1993) [1]

Affinity maps

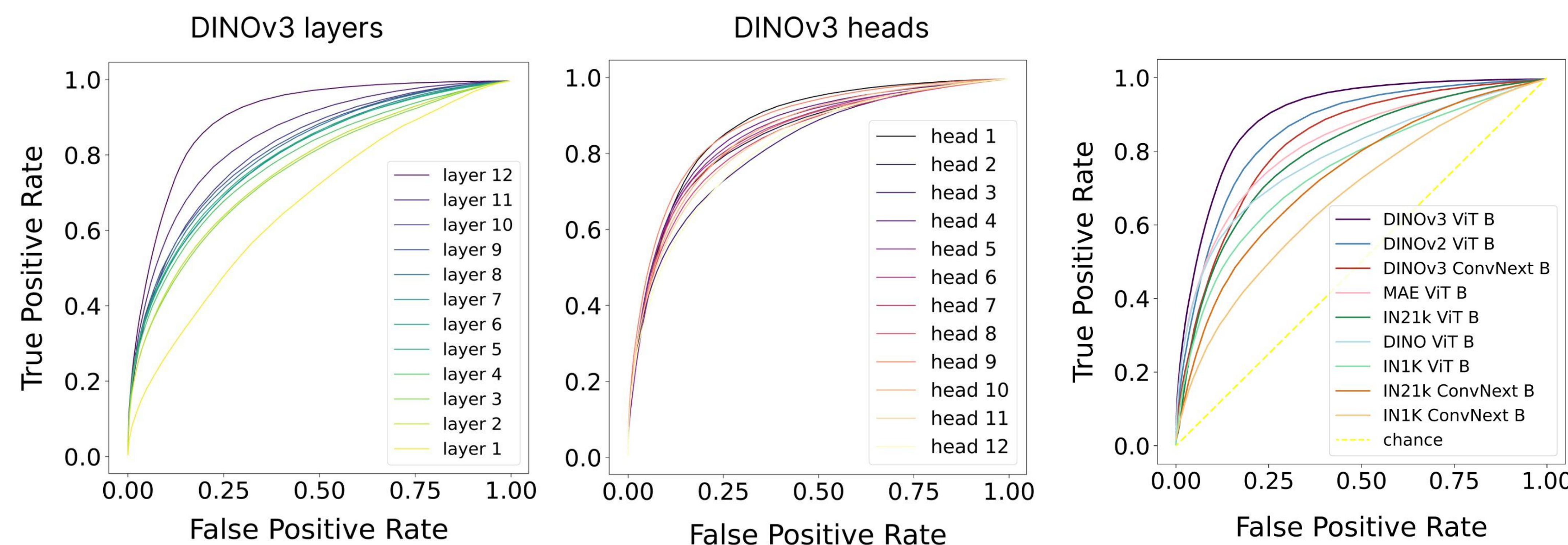
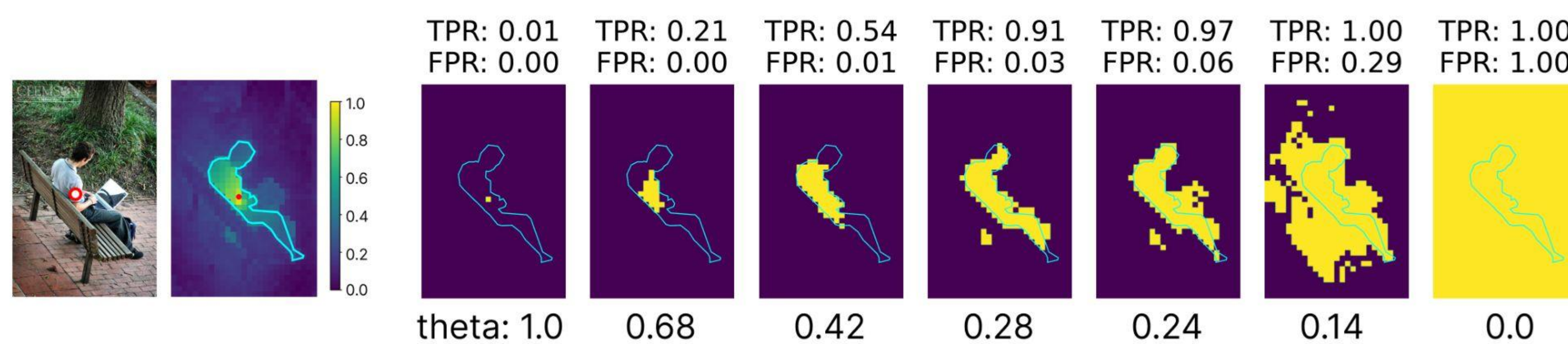


DINOv3 model [2]

- Object grouping in natural scenes requires the visual system to bind image regions into coherent wholes despite ambiguous boundaries and overlapping objects [3].
- Association fields in early visual areas link neurons with similar low-level features, but grouping in natural scenes requires richer, object-centric connectivity in higher visual areas [1,4].
- Here we provide evidence for the existence of such object-centric association fields in higher visual cortex and demonstrate their functional role in perceptual grouping.

Object-centric Representation

For each image, we compute an affinity map by taking the cosine similarity between the patch feature at the central dot location and all other patch features. We then perform an ROC analysis: TPR is the proportion of within-object patches whose affinity exceeds a threshold, FPR is the proportion of outside-object patches. Averaging across all trials yields a summary ROC curve per model, and we use the AUC as our object-centricity metric.

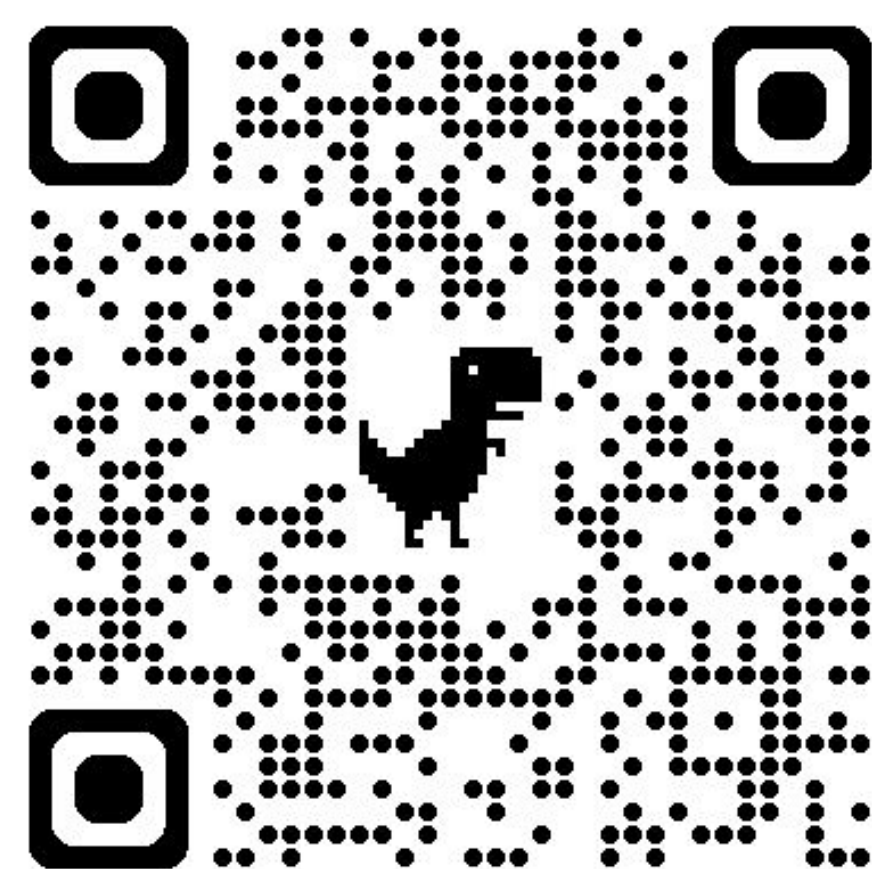


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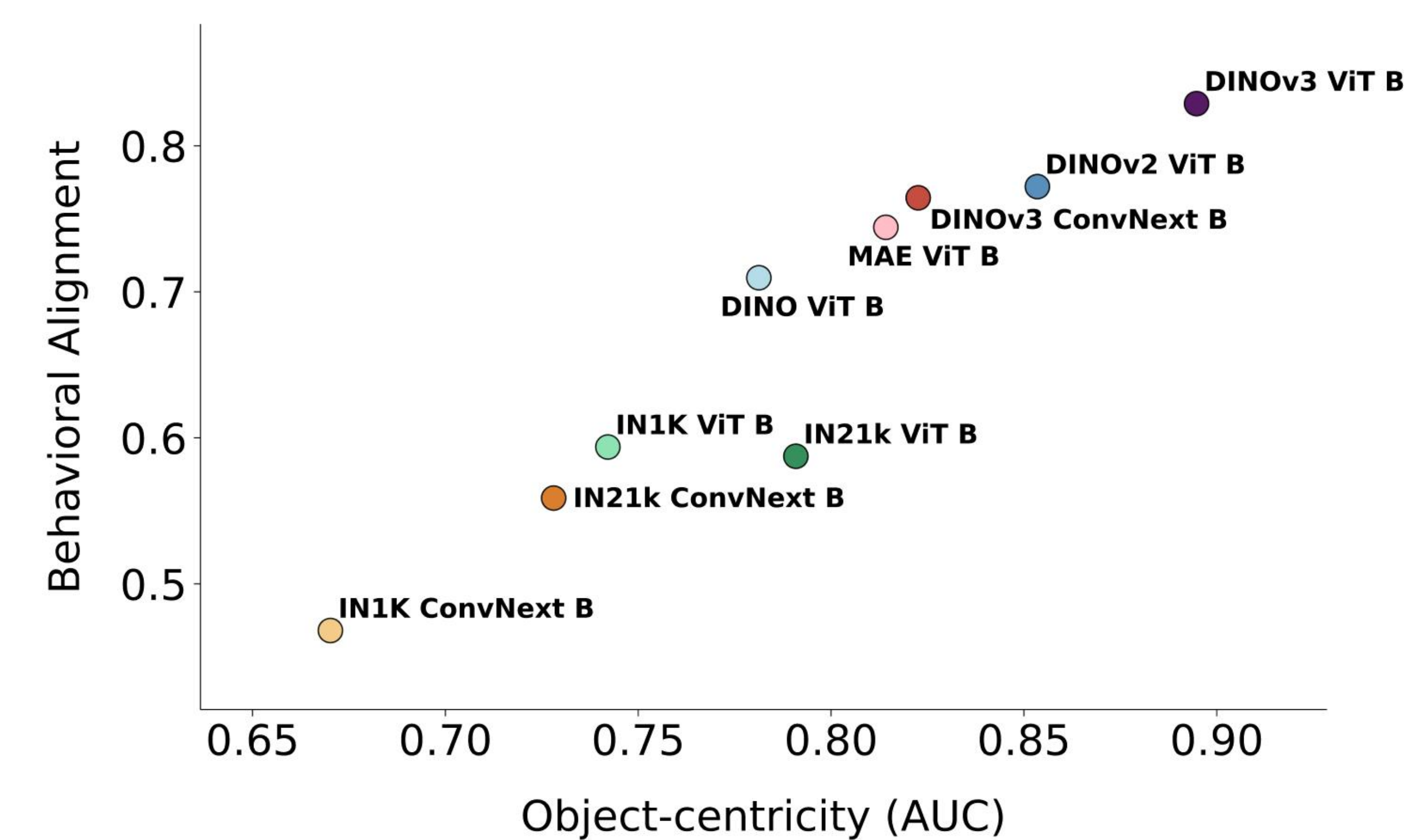
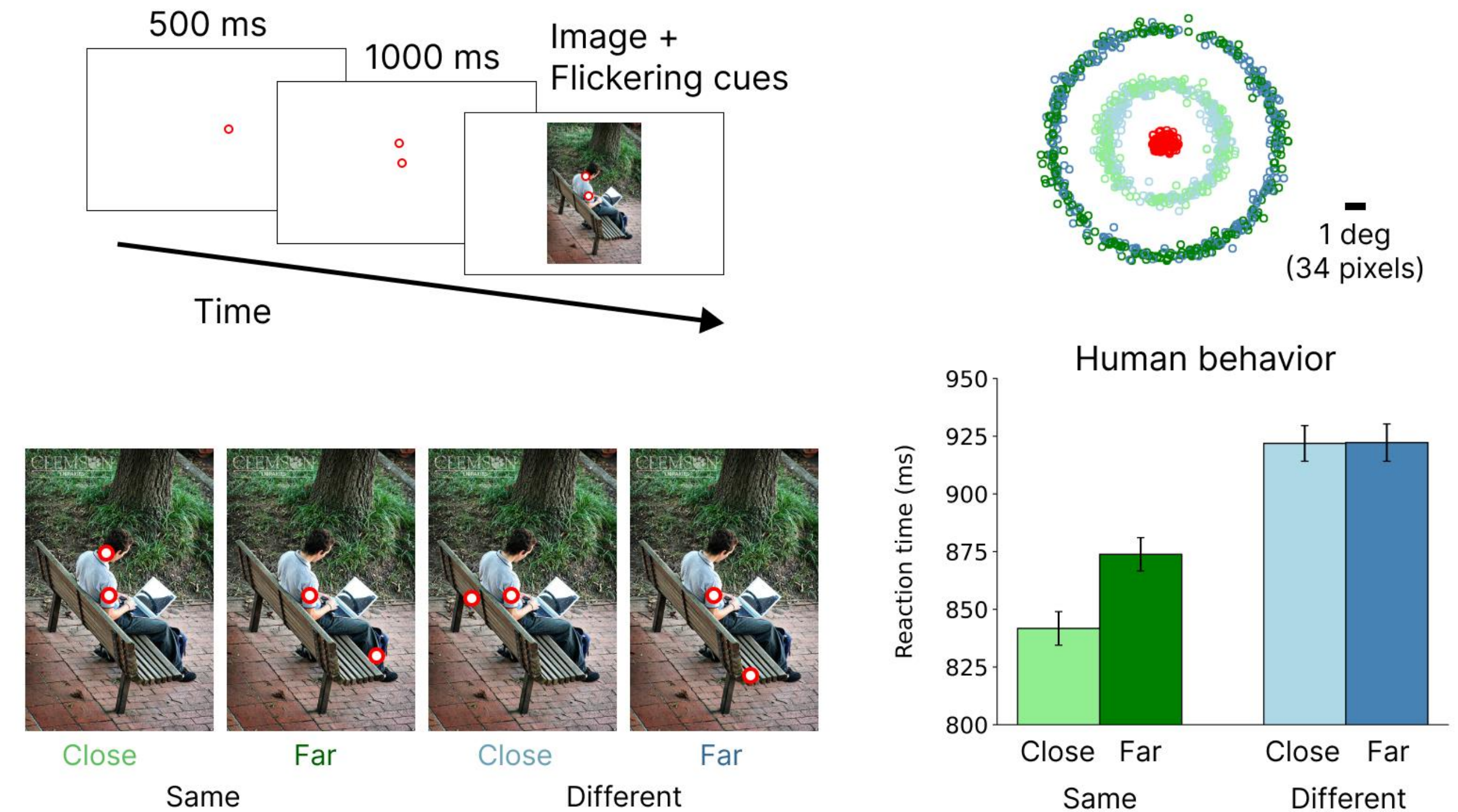
ECCV 2026 paper

CCN Poster



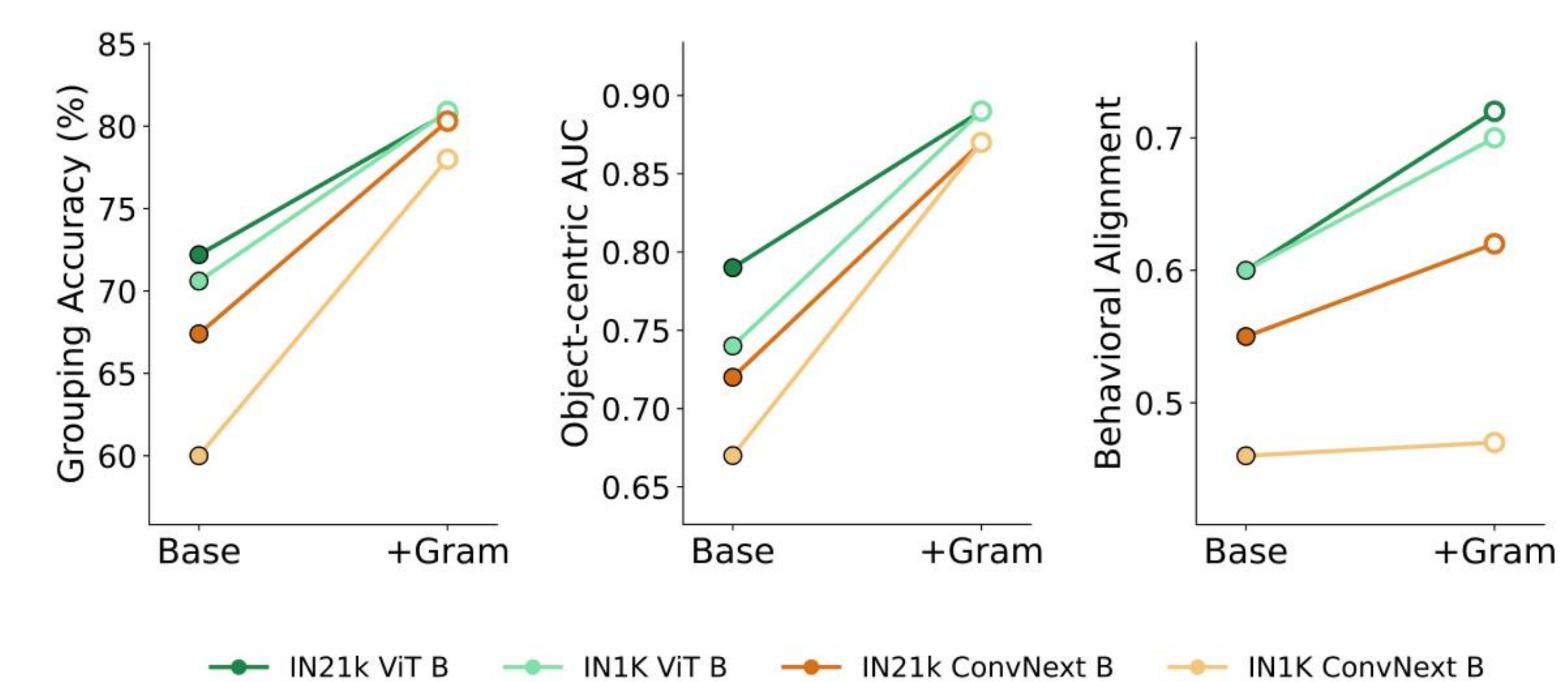
Behavioral evidence

- We collected human judgments using a two-dot paradigm: participants viewed naturalistic scenes and indicated whether two dots were on the same object or different objects.
- 72 participants completed 255 trials each, yielding 1020 trials across 255 images from COCO2017.
- Subjects showed a clear same-object advantage, with faster RTs when both dots were on the same object, and RT increased with dot distance only within the same-object condition.



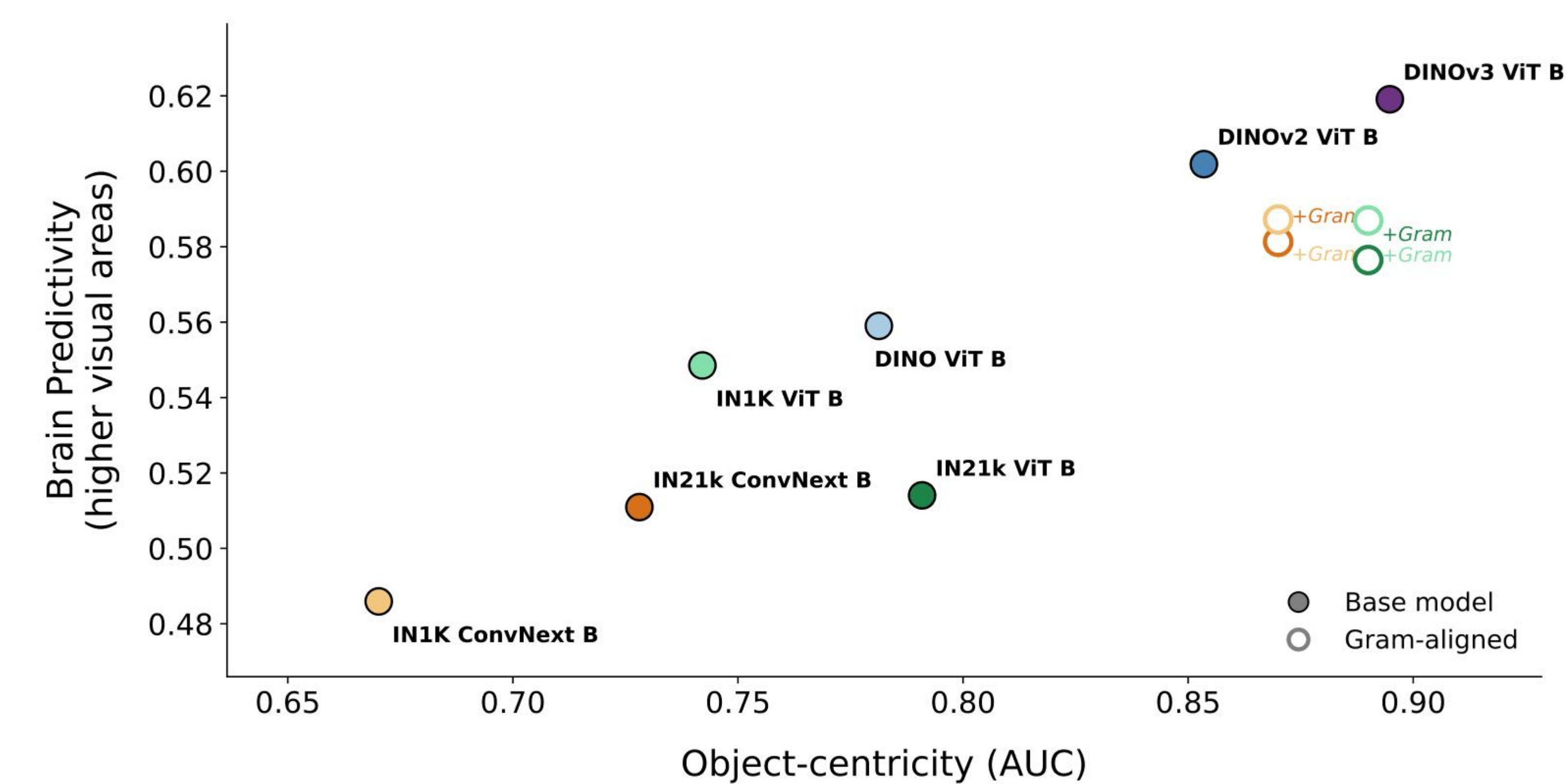
- We fine-tuned supervised models (ViT-B and ConvNext-B) while distilling the Gram matrix structure from DINOv3 ViT-B. The Gram matrix captures the full pairwise patch similarity structure across the image.
- Gram alignment substantially improves grouping accuracy, object-centric AUC, and behavioral alignment across all four models, while a CLS distillation baseline with the same teacher shows no improvement, isolating patch-level correlation structure as the active ingredient.
- Transformer models show larger gains than convolutional ones, likely because self-attention already encodes pairwise patch relationships

Effect of Gram alignment



Neural Evidence

- Using the Natural Scenes Dataset [5], we test whether object-centric structure in model representations predicts brain responses in higher visual areas.
- We use the TBEn encoding model (Adeli et al., 2025) [6] to predict fMRI responses from the full patch feature tensor of each model.
- Models with stronger object-centric structure (higher AUC) better predict neural responses in higher visual areas.



Gram-aligned supervised models show improved brain predictivity relative to their base versions, pointing to object-centric feature correlation structure as the active ingredient.